TMA4212 - Numdiff

Project 1

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1 Introduction

In this project, we analyze a numerical scheme designed to solve reaction-diffusion equations of the form $u_t = \mu u_{xx} + f(u)$ where $\mu > 0$ is the diffusion coefficient, and $f(u)$ represents the reaction term. The reaction term is assumed to be nonstiff, which allows for explicit integration in time. Since diffusion terms generally require implicit methods for stability, we consider a mixed scheme that treats the diffusion implicitly and the reaction explicitly.

We investigate the consistency and stability of a modified Crank-Nicolson scheme. By setting $f(u) = au$ for some constant a, we perform a stability analysis by doing a matrix analysis on the PDE. Our goal is to determine the stability of the method and quantify the global error. To support our theoretical findings, we will conduct numerical experiments.

We will also develop a numerical model to study the spatial spread of an infectious disease using a reaction-diffusion SIR model. This extends the classic SIR model by incorporating diffusion, allowing us to simulate how an outbreak propagates over time and space. The governing equations are

$$\begin{aligned} S_t &= -\beta IS + \mu_s \Delta S \\ I_t &= \beta IS - \gamma I + \mu_I \Delta I \end{aligned}$$

To solve these equations, we will implement a finite difference method, selecting appropriate spatial and temporal discretization schemes. We will simulate an outbreak from an initially localized infection and analyze how different initial conditions influence its spread.

2 Theory

For the model to work numerically we introduce space and time discretization. We let time t be split into N equal timesteps of size $k = 1/N$. For space we let x be split into M equal space steps of size $h = 1/M$. We let the domain of x be $\Omega = [0, 1]$, and t be scaled to $t \in [0, 1]$. Let $x_m = mh$ for all $m \in \{0, .., M\}$ and $t_n = nk$ for all $n \in \{0, .., N\}$.

To solve the reaction-diffusion equation:

$$u_t = \mu u_{xx} + f(u), \tag{1}$$

we will implement a numerical scheme called the modified Crank-Nicolson scheme:

$$U_m^* = U_m^n + \frac{r}{2}(\delta_x^2 U_m^* + \delta_x^2 U_m^n) + kf(U_m^n), \quad r = \mu\frac{k}{h^2}, \tag{2}$$

$$U_m^{n+1} = U_m^* + \frac{k}{2}(f(U_m^*) - f(U_m^n)). \tag{3}$$

Using this, we approximate the solution to (1). We will examine the simple linear case given by:

$$f(u) = au, \quad a \in \mathbb{R} \tag{4}$$

In our scheme the δ_x^2 is defined by:

$$\delta_x^2 U_m = U_{m+1}^n - 2U_i^n + U_{m-1}^n$$

This operation can be represented as a **matrix multiplication**:

$$DU^n = \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \cdots & 0 \\ 0 & 1 & -2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -2 \end{bmatrix} \begin{bmatrix} U_1^n \\ U_2^n \\ U_3^n \\ \vdots \\ U_{M-1}^n \end{bmatrix}$$

2.1 Stability

In order for the scheme to be viable, we require the solution to be stable. To do this, we wish to modify the form of our scheme into:

$$\begin{aligned} U^{n+1} &= CU^n + q^n. \\ U^n &= [U_1^n, U_2^n, ..., U_{M-1}^n]^T. \end{aligned}$$

To check if this scheme is stable, we will analyze the matrix C. In order to deduce how this matrix looks we have to take a look at equation (2) for U^*

$$\begin{aligned} U^* &= U^n + \frac{r}{2}\left(DU^* + DU^n\right) + k \cdot f(U^n) \\ \left(I - \frac{r}{2}D\right)U^* &= \left(I + \frac{r}{2}D\right)U^n + k \cdot f(U^n) \\ U^* &= \left(I - \frac{r}{2}D\right)^{-1} (\left(I + \frac{r}{2}D\right) + k \cdot a \cdot I)U^n \end{aligned}$$

Then by inserting this expression for U^* into (3)

$$\begin{aligned} U^{n+1} &= U^* + \frac{k}{2}(f(U^*) - f(U^n)) \\ U^{n+1} &= \left(I - \frac{r}{2}D\right)^{-1}\left(I + \frac{r}{2}D + kaI\right)U^n + \frac{k}{2}\left(a\left(I - \frac{r}{2}D\right)^{-1}\left(I + \frac{r}{2}D + kaI\right)U^n - aU^n\right) \\ U^{n+1} &= \left(I - \frac{r}{2}D\right)^{-1}\left(I + \frac{r}{2}D + kaI + \frac{k}{2}aI + \frac{k}{2}a\frac{r}{2}D + \frac{k^2a^2}{2}I\right)U^n - \left(I - \frac{r}{2}D\right)^{-1}\left(I - \frac{r}{2}D\right)a\frac{k}{2}U^n \\ U^{n+1} &= \left(I - \frac{r}{2}D\right)^{-1}\left(I + \frac{r}{2}D + akI + \frac{ark}{2}D + \frac{a^2k^2}{2}I\right)U^n \end{aligned}$$

Here it is important to note that D is diagonalizable (and thus invertible) since it is diagonal dominant and irreducible. D can therefore be written as $D = PAP^{-1}$ where P is an orthogonal matrix and A is a diagonal matrix of the eigenvalues of D. Since D is a Tri-diagonal matrix its eigenvalues has the form $\lambda_m = -4\sin^2(\varphi_m)$ with $\varphi_m = \frac{m\pi}{2(M+1)}$ for $m = 1, ..., M$. Also note that $P^{-1} = P^T$

$$\begin{aligned} U^{n+1} &= \left(PP^T - \frac{r}{2}PAP^T\right)^{-1}\left(PP^T + \frac{r}{2}PAP^T + akPP^T + \frac{ark}{2}PAP^T + \frac{a^2k^2}{2}PP^T\right)U^n \\ U^{n+1} &= P\left(I - \frac{r}{2}A\right)^{-1}\left(I + \frac{r}{2}A + akI + \frac{ark}{2}A + \frac{a^2k^2}{2}I\right)P^TU^n \end{aligned}$$

This then gives us a matrix C:

$$C = P\left(I - \frac{r}{2}A\right)^{-1}\left(I + \frac{r}{2}A + akI + \frac{ark}{2}A + \frac{a^2k^2}{2}I\right)P^T$$

Let M $= \left(I - \frac{r}{2}A\right)^{-1}\left(I + \frac{r}{2}A + akI + \frac{ark}{2}A + \frac{a^2k^2}{2}I\right)$. Since M is a product of two diagonal matrices it is also a diagonal matrix .

Since P is invertible and $P^{-1} = P^T$ we can say that

$$\begin{aligned} C &= PMP^T \\ C^T &= \left(PMP^T\right)^T \\ &= \left(P^T\right)^T MP^T \\ &= C \end{aligned}$$

Which means that C is symmetric. Since C is symmetric we are in a special case and the condition for stability is described in BO notes page 55 and says the following:

> If C is symmetric, the condition $\rho(C) \leq 1 + \nu k$ is both necessary and sufficient for stability when using $||\cdot||_{2,h}$.

$$C = P\begin{bmatrix} \frac{1}{1-\frac{r}{2}\lambda_1} & \cdots & 0 \\ \cdots & \cdots & \cdots \\ 0 & \cdots & \frac{1}{1-\frac{r}{2}\lambda_M} \end{bmatrix}\begin{bmatrix} 1+\frac{r}{2}\lambda_1 + ak + \frac{ark}{2}\lambda_1 + \frac{a^2k^2}{2} & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & 1+\frac{r}{2}\lambda_M + ak + \frac{ark}{2}\lambda_M + \frac{a^2k^2}{2} \end{bmatrix}P^T$$

$$C = P\begin{bmatrix} \frac{1+\frac{r}{2}\lambda_1 + ak + \frac{ark}{2}\lambda_1 + \frac{a^2k^2}{2}}{1-\frac{r}{2}\lambda_1} & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & \frac{1+\frac{r}{2}\lambda_M + ak + \frac{ark}{2}\lambda_M + \frac{a^2k^2}{2}}{1-\frac{r}{2}\lambda_M} \end{bmatrix}P^T$$

For calculating the spectral radius i.e the largest eigenvalue we look at elements of the matrix individually

$$\begin{aligned} |\Omega_m| &= \left|\frac{1+\frac{r}{2}\lambda_m + ak + \frac{ark}{2}\lambda_m + \frac{a^2k^2}{2}}{1-\frac{r}{2}\lambda_m}\right| \\ |\Omega_m| &= \left|\frac{1-2r\sin^2(\varphi_m) + ak - 2ark\sin^2(\varphi_m) + \frac{a^2k^2}{2}}{1+2r\sin^2(\varphi_m)}\right| \quad \xRightarrow{\text{triangle inequality}} \\ |\Omega_m| &\leq \left|\frac{1-2r\sin^2(\varphi_m)}{1+2r\sin^2(\varphi_m)}\right| + |ak|\left|\frac{1-2r\sin^2(\varphi_m)}{1+2r\sin^2(\varphi_m)}\right| + \left|\frac{\frac{a^2k^2}{2}}{1+2r\sin^2(\varphi_m)}\right| \\ |\Omega_m| &\leq 1 + |ak| + \left|\frac{a^2k^2}{2}\right| \\ |\Omega_m| &\leq 1 + \left(|a| + \frac{a^2k}{2}\right)k \end{aligned}$$

.

By using the fact that we can choose $k \leq \tau$ where τ is some constant we can say:

$$|\Omega_m| \leq 1 + \left(|a| + \frac{a^2\tau}{2}\right)k$$

.

We have now satisfied our condition and we can conclude that our scheme is unconditionally stable for all r. Which is the desired form. Since our condition for stability is satisfied for all r, the scheme is unconditionally stable.

2.2 Consistency Analysis

For the central difference, we use Taylor expansions and we get:

$$\begin{aligned}
\frac{\delta_x^2 u\left(x_m, t_n\right)}{h^2} &= \frac{u\left(x_{m+1}, t_n\right) - 2u\left(x_m, t_n\right) + u\left(x_{m-1}, t_n\right)}{h^2} \\
\frac{\delta_x^2 u\left(x_m, t_n\right)}{h^2} &= u_{xx}\left(x_m, t_n\right) + \frac{h^2}{12} u_{xxxx}\left(x_m, t_n\right) + O\left(h^4\right)
\end{aligned}$$

The truncation error τ_m is the error made by replace the PDE (1) by the numerical scheme. To find this error, we will apply our scheme to the exact solution. The scheme can be rewritten as :

$$U_m^{n+1} - \frac{r}{2}\delta_x^2 U_m^{n+1} = U_m^n + \frac{r}{2}\delta_x^2 U_m^n + akU_m^n + \frac{ark}{2}\delta_x^2 U_m^n + \frac{a^2k^2}{2}U_m^n$$

We insert the true solution into our equation

$$\begin{aligned}
k\tau_m &= u_m^{n+1} - \frac{r}{2}\delta_x^2 u_m^{n+1} - u_m^n - \frac{r}{2}\delta_x^2 u_m^n - aku_m^n - \frac{ark}{2}\delta_x^2 u_m^n - \frac{a^2k^2}{2}u_m^n \\
k\tau_m &= \left(1 - \frac{r}{2}\delta_x^2\right)\left(u_m^{n+1} - u_m^n\right) - r\delta_x^2 u_m^n - aku_m^n - \frac{ark}{2}\delta_x^2 u_m^n - \frac{a^2k^2}{2}u_m^n
\end{aligned}$$

Expanding this equation we get:

$$\begin{aligned}
k\tau_m =& \left(1 - \frac{\mu k}{2}\left(\partial_x^2 + \frac{1}{12}h^2\partial_x^4 + O(h^4)\right)\right)\left(k\partial_t + \frac{1}{2}k^2\partial_t^2 + \frac{1}{6}k^3\partial_t^3 + O(k^4)\right)u_m^n \\
& - k\mu\left(\partial_x^2 + \frac{1}{12}h^2\partial_x^4 + O(h^4)\right)u_m^n - aku_m^n - \frac{a\mu k^2}{2}\left(\partial_x^2 + \frac{1}{12}h^2\partial_x^4 + O(h^4)\right)u_m^n - \frac{a^2k^2}{2}u_m^n \\
k\tau_m =& \left(k\partial_t + \frac{1}{2}k^2\partial_t^2 + \frac{1}{6}k^3\partial_t^3\right)u_m^n \\
& + \left(-\frac{\mu k^2}{2}\partial_x^2\partial_t - \frac{\mu k^3}{4}\partial_x^2\partial_t^2 - \frac{\mu k^4}{12}\partial_x^2\partial_t^3\right)u_m^n \\
& + \left(-\frac{\mu k^2h^2}{12}\partial_x^4\partial_t - \frac{\mu k^3h^2}{24}\partial_x^4\partial_t^2 - \frac{\mu k^4h^2}{72}\partial_x^4\partial_t^3\right)u_m^n \\
& + \left(-k\mu\partial_x^2 - \frac{\mu kh^2}{12}\partial_x^4 - \frac{a\mu k^2}{2}\partial_x^2 - \frac{a\mu k^2h^2}{12}\partial_x^4\right)u_m^n - aku_m^n - \frac{a^2k^2}{2}u_m^n + O(kh^4 + k^4)
\end{aligned}$$

Knowing that our equation is $u_t = \mu u_{xx} + au \implies \mu u_{xx} = u_t - au$ we can simplify

$$\begin{aligned}
k\tau_m =& \left(\frac{k^3}{6}\partial_t^3 - \frac{\mu k^3}{4}\partial_x^2\partial_t^2 - \frac{\mu k^4}{12}\partial_x^2\partial_t^3\right)u_m^n \\
& + \left(-\frac{\mu k^2h^2}{12}\partial_x^4\partial_t - \frac{\mu k^3h^2}{24}\partial_x^4\partial_t^2 - \frac{\mu k^4h^2}{72}\partial_x^4\partial_t^3\right)u_m^n \\
& + \left(-\frac{\mu kh^2}{12}\partial_x^4 - \frac{a\mu k^2h^2}{12}\partial_x^4\right)u_m^n + O(kh^4 + k^4)
\end{aligned}$$

Since we cannot simplify more we look at the largest polynomials of h and k. Thus we can conclude that the truncation error is $k\tau_m = O\left(k^3 + kh^2\right) \implies \tau_m = O(k^2 + h^2)$

2.3 Convergence:

In order to show convergence we will calculate the global error. Knowing the truncation error this is very straight forward. We should be able to conclude that the global error approaches 0 as our step size for h and k get smaller. The global error is $e^n = u^n - U^n$. We know :

$$\begin{aligned} U^{n+1} = CU^n \text{ and } u^{n+1} = Cu^n + k\tau^n &\implies e^{n+1} = Ce^n + k\tau^n \\ e^n = C^n e^0 + k\sum_{j=0}^{n-1} C^j \tau^{n-j-1} \end{aligned}$$

We make these assumptions:

$$\begin{aligned} e^0 = 0 &\text{ because we choose the initial conditions to be exact } U^0 = u^0 \\ ||C^j|| \leq L \ \forall\ j, \quad L \in \mathbb{R} &\text{ Due to stablilty} \\ ||\tau|| = A(k^2 + h^2), \quad A \in \mathbb{R} & \qquad \left(||\tau|| \xrightarrow[k,h\to 0]{} 0 \right) \end{aligned}$$

Using these assumptions we can find a bound on the global error:

$$\begin{aligned} ||e^n|| &\leq k\sum_{j=0}^{n-1} ||C^j|| ||\tau^{n-j-1}|| \leq knLA\left(h^2 + k^2\right) \\ &= t_n LA\left(h^2 + k^2\right) \\ &\leq TLA\left(h^2 + k^2\right) \xrightarrow[k,h\to 0]{} 0 \end{aligned}$$

This shows us that the error should approach 0 as our step sizes become smaller. We confirm this result in the section 2.4.

2.4 Numerical experiments

Now that we have proven stability and consistency theoretically, we implement our scheme. See project1.ipynb for the implementation as well as additional experiments.

Since our scheme only calculates the values for $U_1, \ldots, U_{M-1}$ we calculate the boundary elements U_0 and U_M by inserting the value for the exact solution in our scheme:

$$U_m^{n+1} - \frac{r}{2}\delta_x^2 U_m^{n+1} = U_m^n + \frac{r}{2}\delta_x^2 U_m^n + akU_m^n + \frac{ark}{2}\delta_x^2 U_m^n + \frac{a^2k^2}{2}U_m^n$$

In order to analyze the error we calculate the numerical solution and compare it with the exact solution.

To do this, we take $u(x,t) = e^{ct}\sin(bx)$. We want to find c and b such u is a solution of (1). We have :

$$\begin{aligned} ce^{ct}\sin(bx) &= -\mu b^2 e^{ct}\sin(bx) + ae^{ct}\sin(bx) \\ c &= -\mu b^2 + a \end{aligned}$$

Where we can choose a, b, μ freely

Below are three different graphs displaying how the true solution compares to the exact solution for some initial data. We also plot the error to show the accuracy of our solution.

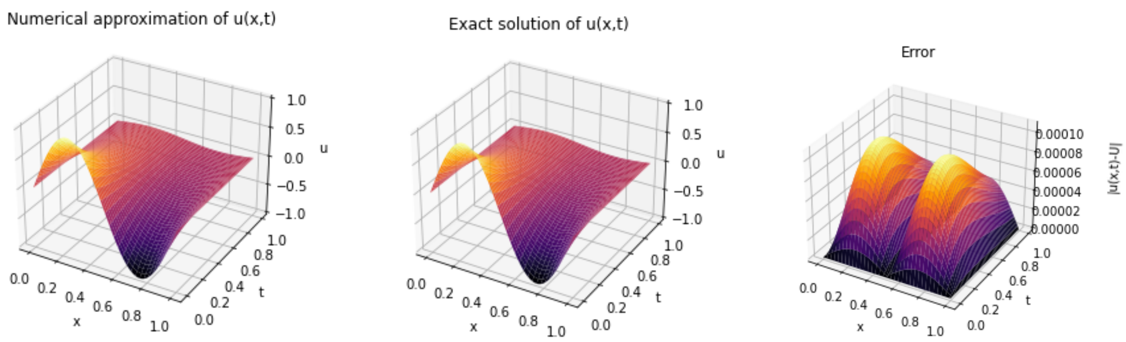


Figure 1: Comparison of the exact solution with the numerical solution with initial data: $a = 1$, $b = 2\pi$, $\mu = 0.1$

We tested our scheme for multiple different step-sizes and values for a, b and μ. The numerical solution behaved nicely for all reasonable values we tested. In the jupyter notebook you can see how the numerical solution behaves for different initial conditions.

From our theoretical results, we know the global error should approach 0 as we decrease our step size. We would like our numerical experiment to show similar results confirming we have implemented the scheme correctly.

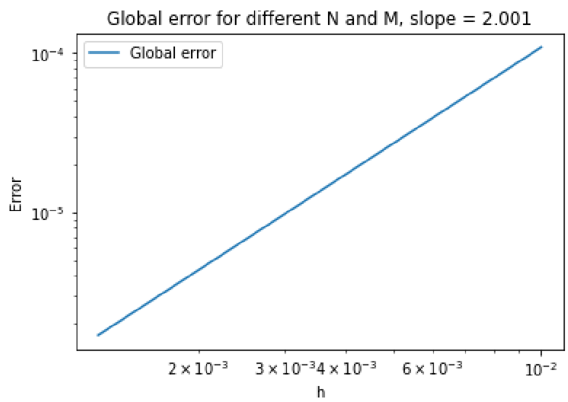


Figure 2: Global error for different sizes of M,N. In the graph h = k at all h.

This figure show, our calculation of the global error decreases when h or k decreases ($||e^n|| \xrightarrow[h,k\to 0]{} 0$). Moreover, the slope is near to two, this is coherent with our theoretical result since $||e||_\infty \leq O(h^2 + k^2)$. This numerical experiment reinforces the idea that our implementation is correct.

3 Application and the SIR model

We wish to investigate how an infectious disease will develop in space and time. We are given a population of N people. The population is divided into 3 classes.

- **Susceptible** S: A person is susceptible if they may get the disease.
- **Infectious** I: A person is infective if they have the disease and are able to transfer it to others.
- **Removed** R: They are a part of the population that may never again infect others or be infected. This class includes the dead, isolated or immune.

We then introduce the system of equations to describe the change in the rate of change in the population:

$$\frac{dS}{dt} = -\beta IS$$
$$\frac{dI}{dt} = \beta IS - \gamma I$$
$$\frac{dR}{dt} = \gamma I$$

Here the constant β symbolizes how fast the people who are susceptible to the disease are infected. If β is large then S(t) will rapidly approach 0 since all **Susceptible** people will become infected quickly. γ symbolizes how fast infectious people get removed from the system. If γ is large then people will quickly go from $I(t)$ to $R(t)$. In the attached python code there are multiple examples of how different values for γ and β influence how the system develops with respect to time.

3.1 Solving the system in a point

In the code, we used an ODE solver to solve the differential equations. Shown in Figure 3 is the simulation for the ODE. We are simulating a population of 10000. Initially we have 1 infected person and the rest are susceptible. Figure 3 shows the percentage of susceptible, infected and removed people at different time intervals. Looking at the graph at time 0 we observe the infected amount rise while susceptible people shrink. At time 5.0 we observe the peak infection. Simultaneously we observe that the number of removed grows at a steady rate. At time 15.0 almost the entire population has been removed from the system and there are very few infected and susceptible people left. We observe that S is monotonically decreasing while R is monotonically increasing. This makes sense as you never add any susceptible people to the population in the model, and once a person is removed (dead) they cant be brought out of the removed state.

3.2 Simulating the problem in space

So far we have only looked at the problem in a single spot in space, we now turn to a spatial model. To incorporate spatial effects, we introduce diffusion terms into the SIR model, resulting in the following system of PDE's:

$$S_t = -\beta IS + \mu_S \Delta S \tag{5}$$
$$I_t = \beta IS - \gamma I + \mu_I \Delta I \tag{6}$$

These equations model the movement of individuals across a continuous space. This means the change of infection is no longer monitored in a single point, but rather dependent on the change of position in 2d (x,y)-axis. The diffusion parameters μ_S and μ_I give the speed at which the infected

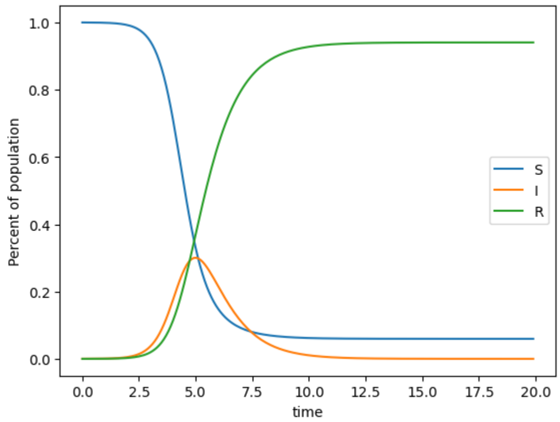


Figure 3: ODE solution with $\gamma = 1$, $\beta = 3$, $I_0 = 1$

and susceptible people move.

For the model to work numerically we reintroduce the space and time discretization. We let time t be split into N equal timesteps of size $k = 1/N$. For space we let both x and y be split into M equal space steps of size $h = 1/M$. We let the domain of x, y be $\Omega = [0,1] \times [0,1]$, and t be scaled to $t \in [0,1]$. Let $x_i = ih$ and $y_j = jh$ for all $i, j \in \{0, .., M\}$ and $t_n = nk$ for all $n \in \{0, .., N\}$. This gives us a natural grid to fit our scheme on. Let $S^n_{i,j}$ be the number of susceptible people at $x = ih$, $y = jh$ and $t = nk$. We use the same notation for the infected ($I^n_{i,j}$).

We use the forward Euler method in time and the central difference in space in order to modify (5) into a scheme that is only dependent on time.

$$\begin{aligned}
\frac{S^{n+1}_{i,j} - S^n_{i,j}}{k} &= -\beta I^n_{i,j} S^n_{i,j} + \mu_s \frac{1}{h^2} \left(S^n_{i-1,j} + S^n_{i+1,j} + S^n_{i,j-1} + S^n_{i,j+1} - 4S^n_{i,j} \right) \\
S^{n+1}_{i,j} &= -k\beta I^n_{i,j} S^n_{i,j} + k\mu_s \frac{1}{h^2} \left(S^n_{i-1,j} + S^n_{i+1,j} + S^n_{i,j-1} + S^n_{i,j+1} - 4S^n_{i,j} \right) + S^n_{i,j} \\
S^{n+1}_{i,j} &= \frac{k\mu_s}{h^2} \left(S^n_{i-1,j} + S^n_{i+1,j} + S^n_{i,j-1} + S^n_{i,j+1} \right) + (1 - k\beta I^n_{i,j} - 4\frac{k\mu_s}{h^2}) S^n_{i,j}
\end{aligned}$$

We then do the same for the infected equation and we find :

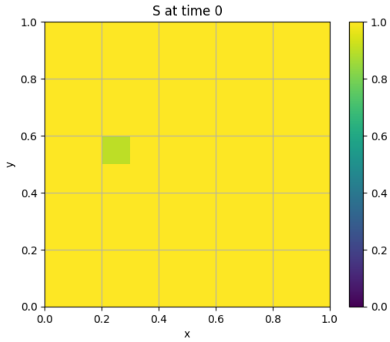
$$I^{n+1}_{i,j} = I^n_{i,j}(k\beta S^n_{i,j} - k\gamma + 1 - 4\frac{k\mu_I}{h^2}) + \frac{k\mu_I}{h^2} \left(I^n_{i-1,j} + I^n_{i+1,j} + I^n_{i,j-1} + I^n_{i,j+1} \right)$$

3.3 Simulating

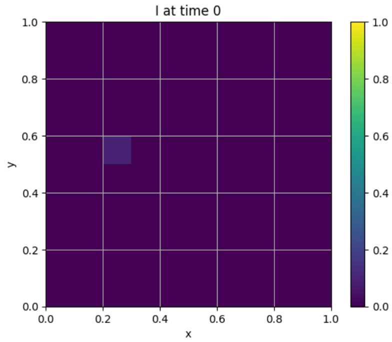
Now that we have a working scheme, we can simulate the spread of infection. We initialize 10% of the population to be infected in the area $i \in [20, 25]$ and $j \in [10, 15]$ and the rest are susceptible.

Shown in Figure 4 are the initial and final states of the susceptible and infected people in the system. Here we chose the parameters $\mu_s = 0,01$, $\mu_i = 0,01$, $\beta = 10$, $\gamma = 3$. The diffusion parameters are chosen to be quite small since our domain is small.

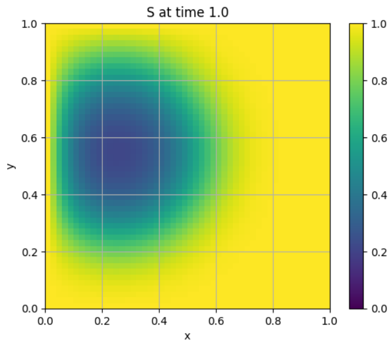
The system behaves differently depending on the initial spread. The jupyter notebook contains multiple visual simulations showing how the infection develops in time. We observe that as time



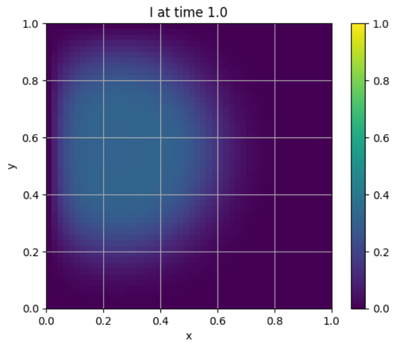
(a) Initial percentage of susceptible people



(b) Initial percentage of infected people



(c) Final percentage of susceptible people



(d) Final percentage of infected people

Figure 4: Initial and final fraction of susceptible and infected people in the model.

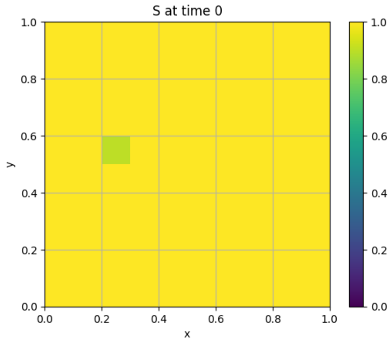
moves forward the percentage of susceptible people in our system decrease, and the number of infected increase. The number of infected is not equal to the number of people removed from susceptible. This is because the system still removes people (the **Removed** plot is shown in the notebook). We also observe that the spread increases uniformly in all directions, until the spread hits the boundary.

Observe in Figure 4c and Figure 4d, that once the spread hits the boundary, the spread slows down and never infects the actual boundary. This occurs by the nature of the scheme, since it takes the solution from the boundary, but never applies to the boundary. That means the boundary effects the spread within the domain. This is shows that the model isn't very robust near the boundary because our choice of boundary affect the spread of the disease. A fix to this would be to increase the resolution of the domain by increasing M. This would reduce the effect the boundary would have on the system. Choosing other boundary conditions could be effective, but that goes beyond the scope of this project.

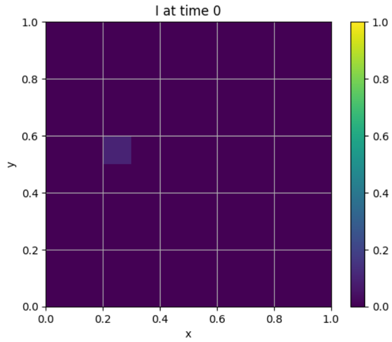
3.4 Simulating people not washing their hands

So far we have assumed that the population in our domain behave the same. This is not true in the real world, so to get a more realistic simulation we have implemented a domain with an area where people dont wash their hands. We modify the scheme such that for $i \in [10, 20]$ and $j \in [30, 40]$, β is chosen to be a value β_{dh}, where $\beta_{dh} > \beta$. This means that the disease will be more infectious in that area. Figure 5 shows the scenario. The parameters and initial spread are the same as for Figure 4, and with $\beta_{dh} = 60$. This means that the disease is six times as infective in the area where people don't wash their hands.

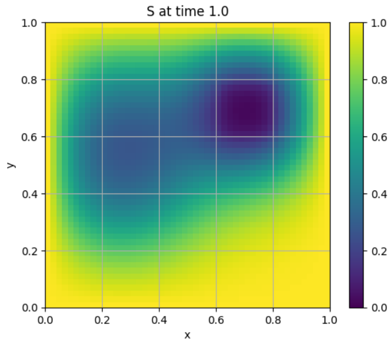
We observed that the initial spread is similar, but once the infection reaches the dirty hands area, the infection explodes into a pandemic, and the final spread is way higher. The gif's in the jupyter



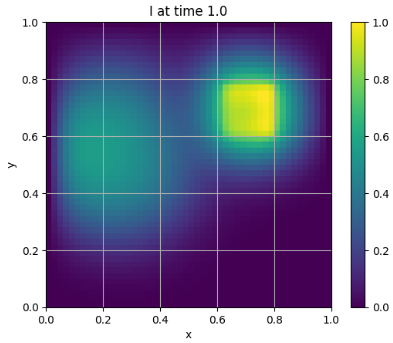
(a) Initial percentage of susceptible people



(b) Initial percentage of infected people



(c) Final percentage of susceptible people



(d) Final percentage of infected people

Figure 5: Simulation on a uniform grid, but with a non uniform β

notebook shows the spread way better as they show the progression over time, and not just the end states. In the notebook we added some systems that are interesting to look at, but we don't have the space to describe in this report. This includes using an infected boundary, instead of a pure susceptible boundary, and utilizing a random initial spread.

4 Conclusion

In this project, we analyzed a numerical scheme for solving reaction-diffusion equations. Through theoretical analysis, we demonstrated that the scheme is unconditionally stable and has a global error that approaches zero as the step sizes decrease. Our numerical experiments confirmed these findings between the theoretical and computed results.

We also created a model of an infectious disease using a reaction-diffusion SIR model that simulated disease propagation over time and space. The model incorporates different parameters for infectiousness and lethality, and for spread in space. It can simulate areas of heightened infection rate, but struggles simulating areas at the boundary of our model. Future work could involve refining boundary conditions and making the model more robust, and studying more effects such as making the disease more lethal to different subjects, or simulating additional conditions on the population.